

Antarang CAP Platform

Final API Specification Document – v1.0

1. Document Purpose

This document defines the final REST API specification for the Antarang Career Assessment Platform MVP.

The API specification supports:

- Login and authentication
- Student, Facilitator/Counsellor, Administrator, and Sub-Administrator roles
- Profile creation and management
- Role-based access control
- Scoped Sub-Administrator access
- Student-to-Facilitator assignment
- Organization hierarchy and cluster management
- Configuration groups
- Platform and assessment language preferences
- Question bank and questionnaire versioning
- Conditional question logic
- Assessment execution
- Count-up timer and elapsed time tracking
- IAR / A-I-A-R-P logic configuration and versioning
- Scoring and evaluation logic versioning
- Benchmarking
- Recommendation generation
- Result templates and report generation
- Counsellor feedback
- Basic external API integration readiness
- Audit and activity logging
- Basic email notification foundation

Data Analyst and Guest User APIs are excluded from MVP and reserved for Phase 2.

2. API Design Standards

Area	Standard
API Style	REST
Base URL	<code>/api/v1</code>
Payload Format	JSON

Area	Standard
Authentication	JWT Bearer Token
Authorization	RBAC + scoped access
URL Naming	Kebab-case
JSON Field Naming	camelCase
ID Format	UUID
Date/Time Format	ISO 8601
Pagination	Required for list APIs
Versioning	URI-based versioning
File Download	API-streamed file or pre-signed S3 URL
Error Format	Standardized error response

3. Common Headers

3.1 Request Headers

Header	Required	Description
<code>Content-Type: application/json</code>	Yes for POST/PUT/PATCH	JSON payload
<code>Authorization: Bearer <token></code>	Yes except login/register APIs	JWT token
<code>X-Tenant-Id</code>	Required for tenant-scoped APIs	Tenant context
<code>Accept-Language</code>	Optional	Preferred response language
<code>X-Request-Id</code>	Optional	Request tracking

4. Standard Response Models

4.1 Success Response

```
{
  "success": true,
  "message": "Request processed successfully",
  "data": {},
}
```

```
"timestamp": "2026-07-01T10:00:00Z"
}
```

4.2 Paginated Response

```
{
  "success": true,
  "message": "Records fetched successfully",
  "data": {
    "content": [],
    "page": 0,
    "size": 20,
    "totalElements": 100,
    "totalPages": 5,
    "last": false
  },
  "timestamp": "2026-07-01T10:00:00Z"
}
```

4.3 Error Response

```
{
  "success": false,
  "errorCode": "VALIDATION_ERROR",
  "message": "Validation failed",
  "details": [
    {
      "field": "email",
      "message": "Invalid email format"
    }
  ],
  "timestamp": "2026-07-01T10:00:00Z",
  "path": "/api/v1/users"
}
```

5. Standard HTTP Status Codes

Status Code	Usage
200	Successful GET, PUT, PATCH
201	Successful creation
204	Successful delete/no content
400	Validation failure/bad request

Status Code	Usage
401	Unauthenticated
403	Unauthorized/insufficient permission
404	Resource not found
409	Duplicate/conflict
422	Business rule validation failure
500	Internal server error

6. Standard Error Codes

Error Code	Description
VALIDATION_ERROR	Request validation failed
AUTH_INVALID_CREDENTIALS	Invalid login credentials
AUTH_TOKEN_EXPIRED	JWT expired
AUTH_TOKEN_INVALID	Invalid token
ACCESS_DENIED	User lacks permission
SCOPE_ACCESS_DENIED	User lacks access to requested scope
RESOURCE_NOT_FOUND	Entity not found
DUPLICATE_RESOURCE	Duplicate record
INVALID_STATE	Action not allowed in current state
CONFIGURATION_NOT_FOUND	Active configuration not found
ASSESSMENT_ALREADY_SUBMITTED	Attempt already submitted
MANDATORY_RESPONSE_MISSING	Mandatory question unanswered
BUSINESS_RULE_FAILED	Business rule validation failed
SYSTEM_ERROR	Unexpected error

7. MVP Roles

7.1 Included in MVP

STUDENT
FACILITATOR

ADMIN
SUB_ADMIN
SUPER_ADMIN

7.2 Phase 2 Roles

DATA_ANALYST
GUEST

8. Authentication APIs

8.1 Register User

Field	Details
Method	POST
Endpoint	/api/v1/auth/register
Auth Required	No
Purpose	Self-registration where enabled

Request

```
{
  "tenantId": "uuid",
  "firstName": "Riya",
  "lastName": "Sharma",
  "email": "riya@example.com",
  "mobileNumber": "9876543210",
  "password": "Password@123",
  "userType": "STUDENT",
  "dateOfBirth": "2010-05-20",
  "primaryOrgUnitId": "uuid",
  "preferredPlatformLanguageId": "uuid",
  "preferredAssessmentLanguageId": "uuid"
}
```

Response

```
{
  "success": true,
  "message": "User registered successfully",
}
```

```
"data": {
  "userId": "uuid",
  "userType": "STUDENT",
  "status": "PENDING",
  "consentRequired": true,
  "consentType": "GUARDIAN"
}
```

8.2 Login

Field	Details
Method	POST
Endpoint	/api/v1/auth/login
Auth Required	No
Purpose	Authenticate user and return JWT

Request

```
{
  "loginId": "riya@example.com",
  "password": "Password@123"
}
```

Response

```
{
  "success": true,
  "message": "Login successful",
  "data": {
    "accessToken": "jwt-token",
    "refreshToken": "refresh-token",
    "expiresIn": 3600,
    "user": {
      "id": "uuid",
      "firstName": "Riya",
      "lastName": "Sharma",
      "userType": "STUDENT",
      "roles": ["STUDENT"],
      "tenantId": "uuid",
      "primaryOrgUnitId": "uuid",
      "preferredPlatformLanguageCode": "hi",
      "preferredAssessmentLanguageCode": "hi"
    }
  }
}
```

```
}  
}
```

8.3 Logout

Field	Details
Method	POST
Endpoint	/api/v1/auth/logout
Auth Required	Yes

Request

```
{  
  "refreshToken": "refresh-token"  
}
```

8.4 Refresh Token

Field	Details
Method	POST
Endpoint	/api/v1/auth/refresh-token
Auth Required	No

Request

```
{  
  "refreshToken": "refresh-token"  
}
```

8.5 Get Logged-in User

Field	Details
Method	GET
Endpoint	/api/v1/auth/me
Auth Required	Yes

Response

```
{
  "success": true,
  "message": "User profile fetched successfully",
  "data": {
    "id": "uuid",
    "firstName": "Riya",
    "lastName": "Sharma",
    "email": "riya@example.com",
    "mobileNumber": "9876543210",
    "userType": "STUDENT",
    "roles": ["STUDENT"],
    "permissions": ["ASSESSMENT_VIEW", "ASSESSMENT_SUBMIT"],
    "tenantId": "uuid",
    "primaryOrgUnitId": "uuid",
    "preferredPlatformLanguageId": "uuid",
    "preferredAssessmentLanguageId": "uuid"
  }
}
```

9. Consent APIs

9.1 Capture Consent

Field	Details
Method	POST
Endpoint	/api/v1/consents
Auth Required	Yes

Request

```
{
  "userId": "uuid",
  "consentType": "GUARDIAN",
  "consentTextVersion": "v1.0",
  "guardianName": "Meena Sharma",
  "guardianContact": "9876543210",
  "consentGiven": true,
  "metadata": {
    "ipAddress": "103.10.10.10",
    "device": "Chrome Web"
  }
}
```



```
}  
}
```

Response

```
{  
  "success": true,  
  "message": "Consent captured successfully",  
  "data": {  
    "consentId": "uuid",  
    "userId": "uuid",  
    "consentType": "GUARDIAN",  
    "consentGivenAt": "2026-07-01T10:00:00Z"  
  }  
}
```

9.2 Get User Consent History

Field	Details
Method	GET
Endpoint	/api/v1/users/{userId}/consents
Auth Required	Yes
Access	Student self, assigned Facilitator, Admin, scoped Sub-Admin

9.3 Withdraw Consent

Field	Details
Method	POST
Endpoint	/api/v1/consents/{consentId}/withdraw
Auth Required	Yes

10. User and Profile Management APIs

10.1 Create User

Field	Details
Method	POST
Endpoint	/api/v1/users
Auth Required	Yes
Access	Admin, scoped Sub-Admin if delegated

Request

```
{
  "tenantId": "uuid",
  "firstName": "Amit",
  "lastName": "Verma",
  "email": "amit@example.com",
  "mobileNumber": "9876543211",
  "userType": "FACILITATOR",
  "primaryOrgUnitId": "uuid",
  "preferredPlatformLanguageId": "uuid",
  "preferredAssessmentLanguageId": "uuid",
  "roleIds": ["uuid"]
}
```

10.2 List Users

Field	Details
Method	GET
Endpoint	/api/v1/users
Auth Required	Yes

Query Parameters

Parameter	Description
userType	STUDENT / FACILITATOR / ADMIN / SUB_ADMIN
orgUnitId	Filter by hierarchy node
status	ACTIVE / INACTIVE
search	Name/email/mobile
page	Page number

Parameter	Description
size	Page size

10.3 Get User by ID

Field	Details
Method	GET
Endpoint	/api/v1/users/{userId}
Auth Required	Yes

10.4 Update User

Field	Details
Method	PUT
Endpoint	/api/v1/users/{userId}
Auth Required	Yes

10.5 Update User Status

Field	Details
Method	PATCH
Endpoint	/api/v1/users/{userId}/status
Auth Required	Yes

Request

```
{
  "status": "INACTIVE",
  "reason": "User no longer part of program"
}
```

10.6 Update Platform Language Preference

Field	Details
Method	PUT
Endpoint	/api/v1/users/{userId}/platform-language
Auth Required	Yes

Request

```
{
  "languageId": "uuid"
}
```

10.7 Update Assessment Language Preference

Field	Details
Method	PUT
Endpoint	/api/v1/users/{userId}/assessment-language
Auth Required	Yes

Request

```
{
  "languageId": "uuid"
}
```

11. RBAC APIs

11.1 List Roles

Field	Details
Method	GET
Endpoint	/api/v1/roles
Auth Required	Yes

11.2 List Permissions

Field	Details
Method	GET
Endpoint	/api/v1/permissions
Auth Required	Yes
Access	Admin

11.3 Assign Roles to User

Field	Details
Method	POST
Endpoint	/api/v1/users/{userId}/roles
Auth Required	Yes
Access	Admin

Request

```
{
  "roles": [
    {
      "roleId": "uuid",
      "scopeType": "ORG_UNIT",
      "scopeId": "uuid"
    }
  ]
}
```

11.4 Remove Role from User

Field	Details
Method	DELETE
Endpoint	/api/v1/users/{userId}/roles/{roleId}
Auth Required	Yes
Access	Admin

12. Organization and Cluster APIs

12.1 Create Organization Unit

Field	Details
Method	POST
Endpoint	/api/v1/org-units
Auth Required	Yes
Access	Admin

Request

```
{
  "tenantId": "uuid",
  "parentOrgUnitId": "uuid",
  "orgUnitType": "INSTITUTION",
  "code": "ABC_SCHOOL",
  "name": "ABC School",
  "description": "Pilot school",
  "address": "Mumbai, Maharashtra",
  "metadata": {
    "board": "State Board"
  }
}
```

12.2 List Organization Units

Field	Details
Method	GET
Endpoint	/api/v1/org-units
Auth Required	Yes

Query Parameters

Parameter	Description
tenantId	Tenant filter
parentOrgUnitId	Parent filter
orgUnitType	STATE / DISTRICT / INSTITUTION / SCHOOL / PROGRAM / COHORT

Parameter	Description
<code>search</code>	Name/code search
<code>page</code>	Page number
<code>size</code>	Page size

12.3 Get Organization Tree

Field	Details
Method	<code>GET</code>
Endpoint	<code>/api/v1/org-units/tree</code>
Auth Required	Yes

12.4 Update Organization Unit

Field	Details
Method	<code>PUT</code>
Endpoint	<code>/api/v1/org-units/{orgUnitId}</code>
Auth Required	Yes

12.5 Create Organizational Cluster

Field	Details
Method	<code>POST</code>
Endpoint	<code>/api/v1/organizational-clusters</code>
Auth Required	Yes
Access	Admin

Request

```
{
  "tenantId": "uuid",
  "code": "MH_PILOT_2026",
  "name": "Maharashtra Pilot 2026",
  "clusterType": "PROGRAM",
}
```

```
"description": "Pilot cluster for 2026"
}
```

12.6 Add Members to Cluster

Field	Details
Method	POST
Endpoint	/api/v1/organizational-clusters/{clusterId}/members
Auth Required	Yes

Request

```
{
  "members": [
    {
      "memberType": "ORG_UNIT",
      "memberId": "uuid"
    }
  ]
}
```

12.7 List Cluster Members

Field	Details
Method	GET
Endpoint	/api/v1/organizational-clusters/{clusterId}/members
Auth Required	Yes

13. Student-Facilitator Assignment APIs

13.1 Assign Students to Facilitator

Field	Details
Method	POST
Endpoint	/api/v1/facilitators/{facilitatorId}/students

Field	Details
Auth Required	Yes
Access	Admin, scoped Sub-Admin

Request

```
{
  "orgUnitId": "uuid",
  "studentIds": ["uuid", "uuid"]
}
```

13.2 Get Facilitator Students

Field	Details
Method	GET
Endpoint	/api/v1/facilitators/{facilitatorId}/students
Auth Required	Yes
Access	Facilitator self, Admin, scoped Sub-Admin

13.3 Unassign Student from Facilitator

Field	Details
Method	DELETE
Endpoint	/api/v1/facilitators/{facilitatorId}/students/{studentId}
Auth Required	Yes
Access	Admin, scoped Sub-Admin

13.4 Get Assignment History

Field	Details
Method	GET
Endpoint	/api/v1/students/{studentId}/assignment-history
Auth Required	Yes

14. Configuration and Language APIs

14.1 Create Configuration Group

Field	Details
Method	POST
Endpoint	/api/v1/configuration-groups
Auth Required	Yes
Access	Admin

Request

```
{
  "tenantId": "uuid",
  "code": "GRADE",
  "name": "Grade",
  "description": "Student grade/class values",
  "isSystemDefined": false
}
```

14.2 List Configuration Groups

Field	Details
Method	GET
Endpoint	/api/v1/configuration-groups
Auth Required	Yes

14.3 Create Configuration Value

Field	Details
Method	POST
Endpoint	/api/v1/configurations
Auth Required	Yes
Access	Admin

Request

```
{
  "configurationGroupId": "uuid",
  "code": "GRADE_10",
  "value": "Grade 10",
  "displayOrder": 1,
  "metadata": {
    "description": "Class 10 students"
  }
}
```

14.4 List Configurations

Field	Details
Method	GET
Endpoint	/api/v1/configurations
Auth Required	Yes

Query Parameters

Parameter	Description
groupCode	Example: GRADE, GENDER
tenantId	Tenant filter

14.5 List Languages

Field	Details
Method	GET
Endpoint	/api/v1/languages
Auth Required	Yes

15. Question Bank APIs

15.1 Create Question Category

Field	Details
Method	POST
Endpoint	/api/v1/question-categories
Auth Required	Yes
Access	Admin

Request

```
{
  "tenantId": "uuid",
  "assessmentType": "INTEREST",
  "code": "ANALYTICAL",
  "name": "Analytical Interest",
  "domainCode": "ANALYTICAL",
  "description": "Analytical interest domain"
}
```

15.2 Create Question

Field	Details
Method	POST
Endpoint	/api/v1/questions
Auth Required	Yes
Access	Admin

Request

```
{
  "tenantId": "uuid",
  "assessmentType": "INTEREST",
  "categoryId": "uuid",
  "questionCode": "INT_Q001",
  "questionType": "RADIO",
  "defaultText": "Which activity do you enjoy the most?",
  "helpText": "Select one option",
  "difficultyLevel": "EASY",
  "reviewStatus": "APPROVED",
  "isRequiredDefault": true,
  "scoringEnabled": true,
}
```

```
"options": [  
  {  
    "optionCode": "A",  
    "defaultText": "Solving puzzles",  
    "scoreValue": 5,  
    "displayOrder": 1  
  },  
  {  
    "optionCode": "B",  
    "defaultText": "Drawing or painting",  
    "scoreValue": 4,  
    "displayOrder": 2  
  }  
]  
}
```

15.3 List Questions

Field	Details
Method	GET
Endpoint	/api/v1/questions
Auth Required	Yes

Query Parameters

Parameter	Description
assessmentType	INTEREST / APTITUDE / REALITY / ASPIRATION
categoryId	Category filter
questionType	RADIO / CHECKBOX / RATING / SHORT_TEXT / LONG_TEXT
reviewStatus	DRAFT / REVIEWED / APPROVED / RETIRED
languageId	Translation availability filter
search	Question text/code search
page	Page number
size	Page size

15.4 Update Question

Field	Details
Method	PUT
Endpoint	/api/v1/questions/{questionId}
Auth Required	Yes
Access	Admin

15.5 Add / Update Question Translations

Field	Details
Method	PUT
Endpoint	/api/v1/questions/{questionId}/translations
Auth Required	Yes

Request

```
{
  "translations": [
    {
      "languageId": "uuid",
      "questionText": "आपको कौन सी गतिविधि सबसे अधिक पसंद है?",
      "helpText": "एक विकल्प चुनें"
    }
  ],
  "optionTranslations": [
    {
      "optionId": "uuid",
      "languageId": "uuid",
      "optionText": "पहेलियाँ हल करना"
    }
  ]
}
```

15.6 Add Question Rule

Field	Details
Method	POST
Endpoint	/api/v1/questions/{questionId}/rules

Field	Details
Auth Required	Yes
Access	Admin

Request

```
{
  "ruleType": "VISIBILITY",
  "ruleConfig": {
    "dependsOnQuestionId": "uuid",
    "operator": "EQUALS",
    "value": "YES",
    "action": "SHOW"
  }
}
```

16. Questionnaire APIs

16.1 Create Questionnaire

Field	Details
Method	POST
Endpoint	/api/v1/questionnaires
Auth Required	Yes
Access	Admin

Request

```
{
  "tenantId": "uuid",
  "assessmentType": "INTEREST",
  "code": "INTEREST_QUESTIONNAIRE",
  "name": "Interest Assessment Questionnaire",
  "description": "Interest assessment questionnaire",
  "shuffleQuestions": false
}
```

16.2 Create Questionnaire Version

Field	Details
Method	POST
Endpoint	/api/v1/questionnaires/{questionnaireId}/versions
Auth Required	Yes

Request

```
{
  "versionNotes": "Initial draft version"
}
```

16.3 Add Questions to Questionnaire Version

Field	Details
Method	POST
Endpoint	/api/v1/questionnaire-versions/{versionId}/questions
Auth Required	Yes

Request

```
{
  "questions": [
    {
      "questionId": "uuid",
      "displayOrder": 1,
      "isMandatory": true,
      "sectionCode": "SECTION_1",
      "sectionName": "Interest Discovery",
      "conditionConfig": null
    }
  ]
}
```

16.4 Reorder Questionnaire Questions

Field	Details
Method	PUT

Field	Details
Endpoint	/api/v1/questionnaire-versions/{versionId}/questions/reorder
Auth Required	Yes

16.5 Preview Questionnaire Version

Field	Details
Method	GET
Endpoint	/api/v1/questionnaire-versions/{versionId}/preview
Auth Required	Yes

16.6 Publish Questionnaire Version

Field	Details
Method	POST
Endpoint	/api/v1/questionnaire-versions/{versionId}/publish
Auth Required	Yes
Access	Admin

Request

```
{
  "versionNotes": "Approved for MVP release"
}
```

16.7 List Questionnaire Versions

Field	Details
Method	GET
Endpoint	/api/v1/questionnaires/{questionnaireId}/versions
Auth Required	Yes

17. Assessment Configuration APIs

17.1 Create Assessment

Field	Details
Method	POST
Endpoint	/api/v1/assessments
Auth Required	Yes
Access	Admin

Request

```
{
  "tenantId": "uuid",
  "code": "INTEREST",
  "name": "Interest Assessment",
  "description": "Interest assessment"
}
```

17.2 Create Assessment Version

Field	Details
Method	POST
Endpoint	/api/v1/assessments/{assessmentId}/versions
Auth Required	Yes

Request

```
{
  "questionnaireVersionId": "uuid",
  "versionNumber": 1
}
```

17.3 Configure Assessment Execution Rules

Field	Details
Method	POST
Endpoint	/api/v1/assessment-versions/{versionId}/configuration
Auth Required	Yes

Request

```
{
  "timerMode": "COUNT_UP",
  "maxDurationMinutes": null,
  "autoSubmitEnabled": false,
  "showTimerToStudent": true,
  "allowResume": true,
  "restartOnInterruption": false,
  "allowReattempt": true,
  "reattemptAfterDays": 365,
  "maxAttempts": 1,
  "completionRule": "ALL_MANDATORY"
}
```

18. Assessment Configuration Group APIs

18.1 Create Configuration Group

Field	Details
Method	POST
Endpoint	/api/v1/assessment-configuration-groups
Auth Required	Yes
Access	Admin

Request

```
{
  "tenantId": "uuid",
  "code": "MH_GRADE10_2026",
  "name": "Maharashtra Grade 10 Assessment Group",
  "description": "Assessment configuration for Maharashtra Grade 10",
  "academicYear": "2026-27",
}
```

```
"assessmentCycle": "Cycle 1"
}
```

18.2 Add Configuration Group Items

Field	Details
Method	POST
Endpoint	/api/v1/assessment-configuration-groups/{groupId}/items
Auth Required	Yes
Access	Admin

Request

```
{
  "items": [
    {
      "assessmentId": "uuid",
      "assessmentVersionId": "uuid",
      "questionnaireVersionId": "uuid",
      "iarLogicVersionId": "uuid",
      "scoringRuleVersionId": "uuid",
      "assessmentLanguageId": "uuid",
      "isRequired": true,
      "displayOrder": 1
    }
  ]
}
```

18.3 Configure Group Output

Field	Details
Method	POST
Endpoint	/api/v1/assessment-configuration-groups/{groupId}/outputs
Auth Required	Yes

Request

```
{
  "reportTemplateVersionId": "uuid",
  "outputLanguageId": "uuid",
}
```

```

"outputConfig": {
  "showScores": true,
  "showRecommendations": true,
  "showCounsellorFeedback": true
}
}

```

18.4 Assign Configuration Group

Field	Details
Method	POST
Endpoint	/api/v1/assessment-configuration-groups/{groupId}/assignments
Auth Required	Yes
Access	Admin, scoped Sub-Admin where permitted

Request

```

{
  "assignmentType": "ORG_UNIT",
  "assignmentId": "uuid",
  "effectiveFrom": "2026-07-01T00:00:00Z",
  "effectiveTo": "2027-03-31T23:59:59Z"
}

```

18.5 Activate Configuration Group

Field	Details
Method	POST
Endpoint	/api/v1/assessment-configuration-groups/{groupId}/activate
Auth Required	Yes
Access	Admin

18.6 Get Active Configuration for Student

Field	Details
Method	GET

Field	Details
Endpoint	/api/v1/students/{studentId}/active-assessment-configuration
Auth Required	Yes
Access	Student self, assigned Facilitator, Admin, scoped Sub-Admin

19. Assessment Execution APIs

19.1 List Available Assessments for Student

Field	Details
Method	GET
Endpoint	/api/v1/students/{studentId}/assessments
Auth Required	Yes

Response

```
{
  "success": true,
  "message": "Assessments fetched successfully",
  "data": [
    {
      "assessmentId": "uuid",
      "assessmentCode": "INTEREST",
      "assessmentName": "Interest Assessment",
      "configurationGroupId": "uuid",
      "status": "NOT_STARTED",
      "timerMode": "COUNT_UP",
      "allowResume": true
    }
  ]
}
```

19.2 Start Assessment

Field	Details
Method	POST
Endpoint	/api/v1/assessments/{assessmentId}/start

Field	Details
Auth Required	Yes
Access	Student, Facilitator/Admin for demo/testing

Request

```
{
  "studentId": "uuid",
  "assessmentLanguageId": "uuid"
}
```

Response

```
{
  "success": true,
  "message": "Assessment started successfully",
  "data": {
    "attemptId": "uuid",
    "assessmentId": "uuid",
    "configurationGroupId": "uuid",
    "questionnaireVersionId": "uuid",
    "iarLogicVersionId": "uuid",
    "scoringRuleVersionId": "uuid",
    "reportTemplateVersionId": "uuid",
    "assessmentLanguageId": "uuid",
    "timer": {
      "timerMode": "COUNT_UP",
      "startedAt": "2026-07-01T10:00:00Z",
      "showTimerToStudent": true,
      "autoSubmitEnabled": false
    },
    "questions": [
      {
        "questionId": "uuid",
        "questionType": "RADIO",
        "questionText": "Which activity do you enjoy the most?",
        "isMandatory": true,
        "displayOrder": 1,
        "options": [
          {
            "optionId": "uuid",
            "optionText": "Solving puzzles",
            "displayOrder": 1
          }
        ]
      }
    ],
    "conditionConfig": null
  }
}
```

```
}  
}
```

19.3 Save Responses

Field	Details
Method	PUT
Endpoint	/api/v1/attempts/{attemptId}/responses
Auth Required	Yes
Access	Attempt owner

Request

```
{  
  "responses": [  
    {  
      "questionId": "uuid",  
      "selectedOptionIds": ["uuid"],  
      "responseText": null,  
      "responseNumeric": null,  
      "isSkipped": false  
    }  
  ]  
}
```

19.4 Save Question Timing

Field	Details
Method	PUT
Endpoint	/api/v1/attempts/{attemptId}/question-timings
Auth Required	Yes

Request

```
{  
  "questionId": "uuid",  
  "startedAt": "2026-07-01T10:02:00Z",  
  "endedAt": "2026-07-01T10:03:30Z",  
}
```



```
"elapsedSeconds": 90
}
```

19.5 Submit Assessment

Field	Details
Method	POST
Endpoint	/api/v1/attempts/{attemptId}/submit
Auth Required	Yes

Response

```
{
  "success": true,
  "message": "Assessment submitted successfully",
  "data": {
    "attemptId": "uuid",
    "status": "SUBMITTED",
    "startedAt": "2026-07-01T10:00:00Z",
    "submittedAt": "2026-07-01T10:42:30Z",
    "elapsedSeconds": 2550,
    "submissionReason": "MANUAL_SUBMIT",
    "scoreStatus": "PENDING"
  }
}
```

19.6 Attempt History

Field	Details
Method	GET
Endpoint	/api/v1/students/{studentId}/attempt-history
Auth Required	Yes

20. Scoring and Benchmark APIs

20.1 Create Scoring Rule

Field	Details
Method	POST
Endpoint	/api/v1/scoring-rules
Auth Required	Yes
Access	Admin

20.2 Create Scoring Rule Version

Field	Details
Method	POST
Endpoint	/api/v1/scoring-rules/{scoringRuleId}/versions
Auth Required	Yes

Request

```
{
  "versionNumber": 1,
  "ruleDefinition": {
    "domains": [
      {
        "domainCode": "ANALYTICAL",
        "questionCodes": ["INT_Q001", "INT_Q002"],
        "aggregation": "SUM"
      }
    ]
  },
  "versionNotes": "Initial scoring version"
}
```

20.3 Publish Scoring Rule Version

Field	Details
Method	POST
Endpoint	/api/v1/scoring-rule-versions/{versionId}/publish
Auth Required	Yes

20.4 Calculate Score

Field	Details
Method	POST
Endpoint	/api/v1/scoring/calculate/{attemptId}
Auth Required	Yes
Access	System/Admin

Response

```
{
  "success": true,
  "message": "Score calculated successfully",
  "data": {
    "attemptId": "uuid",
    "scores": [
      {
        "domainCode": "ANALYTICAL",
        "rawScore": 72,
        "normalizedScore": 72,
        "benchmarkLabel": "HIGH",
        "interpretation": "Strong analytical interest"
      }
    ]
  }
}
```

20.5 Create Benchmark

Field	Details
Method	POST
Endpoint	/api/v1/benchmarks
Auth Required	Yes
Access	Admin

Request

```
{
  "tenantId": "uuid",
  "assessmentId": "uuid",
  "domainCode": "ANALYTICAL",
```

```
"gradeConfigId": "uuid",
"ageGroupConfigId": "uuid",
"minScore": 71,
"maxScore": 100,
"benchmarkLabel": "HIGH",
"interpretation": "Strong analytical inclination"
}
```

20.6 Get Student Scores

Field	Details
Method	GET
Endpoint	/api/v1/students/{studentId}/scores
Auth Required	Yes

21. IAR / A-I-A-R-P Logic APIs

21.1 Create IAR Logic Config

Field	Details
Method	POST
Endpoint	/api/v1/iar-logic-configs
Auth Required	Yes
Access	Admin

Request

```
{
  "tenantId": "uuid",
  "code": "DEFAULT_IAR_2026",
  "name": "Default IAR Logic 2026",
  "description": "Default IAR recommendation logic"
}
```

21.2 Create IAR Logic Version

Field	Details
Method	POST
Endpoint	/api/v1/iar-logic-configs/{logicConfigId}/versions
Auth Required	Yes

Request

```
{
  "versionNumber": 1,
  "logicDefinition": {
    "aspirationWeight": 0.25,
    "interestWeight": 0.25,
    "aptitudeWeight": 0.30,
    "realityWeight": 0.20,
    "rules": []
  },
  "versionNotes": "Initial IAR logic version"
}
```

21.3 Validate IAR Logic Version

Field	Details
Method	POST
Endpoint	/api/v1/iar-logic-versions/{versionId}/validate
Auth Required	Yes

21.4 Publish IAR Logic Version

Field	Details
Method	POST
Endpoint	/api/v1/iar-logic-versions/{versionId}/publish
Auth Required	Yes

22. Career and Recommendation APIs

22.1 Create Career Cluster

Field	Details
Method	POST
Endpoint	/api/v1/career-clusters
Auth Required	Yes
Access	Admin

22.2 Create Career

Field	Details
Method	POST
Endpoint	/api/v1/careers
Auth Required	Yes
Access	Admin

22.3 Create Career Mapping

Field	Details
Method	POST
Endpoint	/api/v1/career-mappings
Auth Required	Yes
Access	Admin

22.4 Generate Recommendation

Field	Details
Method	POST
Endpoint	/api/v1/recommendations/generate/{studentId}
Auth Required	Yes

Field	Details
Access	System/Admin

Response

```
{
  "success": true,
  "message": "Recommendations generated successfully",
  "data": {
    "recommendationRunId": "uuid",
    "studentId": "uuid",
    "recommendations": [
      {
        "careerId": "uuid",
        "careerName": "Data Analyst",
        "rankOrder": 1,
        "recommendationScore": 88,
        "reason": "Strong analytical interest and numerical aptitude",
        "matchBreakdown": {
          "interest": "HIGH",
          "aptitude": "HIGH",
          "reality": "MEDIUM",
          "aspiration": "MATCH"
        }
      }
    ]
  }
}
```

22.5 Get Student Recommendations

Field	Details
Method	GET
Endpoint	/api/v1/students/{studentId}/recommendations
Auth Required	Yes

23. Report Template and Report APIs

23.1 Create Report Template

Field	Details
Method	POST
Endpoint	/api/v1/report-templates
Auth Required	Yes
Access	Admin

23.2 Create Report Template Version

Field	Details
Method	POST
Endpoint	/api/v1/report-templates/{templateId}/versions
Auth Required	Yes

23.3 Publish Report Template Version

Field	Details
Method	POST
Endpoint	/api/v1/report-template-versions/{versionId}/publish
Auth Required	Yes

23.4 Generate Report

Field	Details
Method	POST
Endpoint	/api/v1/reports/generate/{studentId}
Auth Required	Yes
Access	System/Admin

Response

```
{  
  "success": true,
```



```

"message": "Report generated successfully",
"data": {
  "reportId": "uuid",
  "studentId": "uuid",
  "fileName": "career-report-riya-sharma.pdf",
  "downloadUrl": "/api/v1/reports/uuid/download"
}
}

```

23.5 Get Student Reports

Field	Details
Method	GET
Endpoint	/api/v1/students/{studentId}/reports
Auth Required	Yes

23.6 Download Report

Field	Details
Method	GET
Endpoint	/api/v1/reports/{reportId}/download
Auth Required	Yes

Response: PDF file stream or pre-signed download URL.

24. Counsellor Feedback APIs

24.1 Add Feedback

Field	Details
Method	POST
Endpoint	/api/v1/students/{studentId}/feedback
Auth Required	Yes
Access	Assigned Facilitator, Admin

Request

```
{
  "reportId": "uuid",
  "feedbackText": "Student shows strong analytical inclination.",
  "recommendationNotes":
    "Discuss data analytics and commerce pathway in next counselling session.",
  "visibility": "VISIBLE_TO_STUDENT"
}
```

24.2 Get Student Feedback

Field	Details
Method	GET
Endpoint	/api/v1/students/{studentId}/feedback
Auth Required	Yes
Access	Student self, assigned Facilitator, Admin, scoped Sub-Admin

24.3 Update Feedback

Field	Details
Method	PUT
Endpoint	/api/v1/feedback/{feedbackId}
Auth Required	Yes
Access	Feedback author or Admin

25. Dashboard and Analytics APIs

25.1 Student Dashboard

Field	Details
Method	GET
Endpoint	/api/v1/students/{studentId}/dashboard
Auth Required	Yes

Field	Details
Access	Student self

25.2 Facilitator Dashboard

Field	Details
Method	GET
Endpoint	/api/v1/facilitators/{facilitatorId}/dashboard
Auth Required	Yes
Access	Facilitator self, Admin

25.3 Admin Dashboard

Field	Details
Method	GET
Endpoint	/api/v1/admin/dashboard
Auth Required	Yes
Access	Admin

Query Parameters

Parameter	Description
tenantId	Tenant filter
orgUnitId	Org scope filter
clusterId	Cluster filter
fromDate	Start date
toDate	End date

25.4 Sub-Admin Dashboard

Field	Details
Method	GET
Endpoint	/api/v1/sub-admin/dashboard

Field	Details
Auth Required	Yes
Access	Sub-Admin scoped

26. Notification APIs

26.1 Send Notification

Field	Details
Method	POST
Endpoint	/api/v1/notifications/send
Auth Required	Yes
Access	System/Admin

Request

```
{
  "userId": "uuid",
  "templateCode": "ACCOUNT_CREATED",
  "channel": "EMAIL",
  "variables": {
    "firstName": "Riya",
    "loginUrl": "https://cap.antarang.org"
  }
}
```

26.2 List Notification Logs

Field	Details
Method	GET
Endpoint	/api/v1/notification-logs
Auth Required	Yes
Access	Admin

27. External Integration APIs

These APIs provide basic MVP integration readiness. Full Salesforce/Glific connectors are Phase 2.

27.1 Register Integration Client

Field	Details
Method	POST
Endpoint	/api/v1/integration-clients
Auth Required	Yes
Access	Admin

27.2 Create / Upsert Student from External System

Field	Details
Method	POST
Endpoint	/api/v1/integrations/students
Auth Required	API key / integration token

Request

```
{
  "externalSystem": "SALESFORCE",
  "externalStudentId": "SF_12345",
  "tenantId": "uuid",
  "firstName": "Riya",
  "lastName": "Sharma",
  "mobileNumber": "9876543210",
  "email": "riya@example.com",
  "primaryOrgUnitId": "uuid",
  "grade": "GRADE_10"
}
```

27.3 Get Student Result for External System

Field	Details
Method	GET

Field	Details
Endpoint	/api/v1/integrations/students/{externalStudentId}/results
Auth Required	API key / integration token

27.4 Get Student Report for External System

Field	Details
Method	GET
Endpoint	/api/v1/integrations/students/{externalStudentId}/reports/latest
Auth Required	API key / integration token

27.5 List Integration Events

Field	Details
Method	GET
Endpoint	/api/v1/integration-events
Auth Required	Yes
Access	Admin

28. Audit and Activity APIs

28.1 List Audit Logs

Field	Details
Method	GET
Endpoint	/api/v1/audit-logs
Auth Required	Yes
Access	Admin, limited scoped Sub-Admin where applicable

Query Parameters

Parameter	Description
<code>entityName</code>	Entity/table name
<code>entityId</code>	Entity ID
<code>performedBy</code>	User ID
<code>fromDate</code>	Start date
<code>toDate</code>	End date

28.2 List Activity Logs

Field	Details
Method	<code>GET</code>
Endpoint	<code>/api/v1/activity-logs</code>
Auth Required	Yes
Access	Admin

29. Authorization Matrix

Capability	Student	Facilitator	Admin	Sub-Admin
Login	Yes	Yes	Yes	Yes
Own Profile	Yes	Yes	Yes	Yes
Update Own Language	Yes	Yes	Yes	Yes
Take Assessment	Yes	Demo/Training	Demo/ Testing	No
View Own Results	Yes	Yes, demo only	Yes	No
View Assigned Students	No	Yes	Yes	Scoped
Create Users	No	Limited if delegated	Yes	Limited if delegated
Manage Roles	No	No	Yes	No
Assign Student to Facilitator	No	No	Yes	Scoped Yes
Manage Org Hierarchy	No	No	Yes	No

Capability	Student	Facilitator	Admin	Sub-Admin
Manage Clusters	No	No	Yes	Limited if delegated
Manage Question Bank	No	Delegated only	Yes	No
Manage Questionnaires	No	Delegated only	Yes	No
Manage IAR Logic	No	No	Yes	No
Manage Scoring Logic	No	No	Yes	No
Manage Configuration Groups	No	No	Yes	No
Select Output Template	No	No	Yes	Scoped Yes
Configure Output Language	No	No	Yes	Scoped Yes
Add Counsellor Feedback	No	Assigned students	Yes	No
View Analytics	Own only	Assigned cohort	Full	Scoped
Export Reports	Own only	Assigned cohort	Full	Scoped
View Audit Logs	No	No	Yes	Scoped limited

30. MVP API Scope by Sprint

Sprint 1 APIs

API Area	Scope
Auth	Register, login, logout, refresh, me
Users	CRUD, status update, profile
RBAC	Roles, permissions, role assignment
Consent	Capture and view consent
Organization	Org unit CRUD, org tree
Assignment	Student-facilitator mapping
Config	Configuration groups/values
Language	List/update platform and assessment language

Sprint 2 APIs

API Area	Scope
Question Bank	Question category, questions, options, translations
Questionnaire	Questionnaire, versions, add questions, publish
Assessment Config	Assessment, assessment version, timer/config
Configuration Group	Create group, add items/output, assign group
IAR Config	Create IAR config/version

Sprint 3 APIs

API Area	Scope
Assessment Execution	List, start, save, timing, submit
Scoring	Scoring rules, scoring version, calculate score
Benchmark	Create/list benchmarks
Attempt History	Student attempts and statuses
Traceability	Attempt linked to configuration versions

Sprint 4 APIs

API Area	Scope
Recommendation	Generate/list recommendations
Reports	Templates, generate/download reports
Feedback	Add/list/update counsellor feedback
Dashboard	Student, facilitator, admin, sub-admin dashboards
Notifications	Send/log email notifications
Integration	Basic external student/result/report APIs
Audit	Audit/activity logs

31. Phase 2 API Scope

Phase 2 Area	API Scope
Data Analyst Role	Analytics views, dataset export, advanced dashboards
Guest User Role	Demo login, demo assessments, sample dashboards

Phase 2 Area	API Scope
SMS Notifications	SMS provider, retry, delivery reports
Salesforce/Glific Connectors	Production connector-specific APIs
Advanced Rule Builder	Rule simulation, validation, approval workflow
Advanced Reporting	Custom reports, scheduled exports

32. Final API Design Notes

1. The API design supports scoped RBAC for Sub-Admin access.
2. Guest and Data Analyst APIs are intentionally excluded from MVP.
3. Assessment configuration groups are treated as a first-class API module.
4. Every assessment attempt must capture configuration/version traceability.
5. Platform language and assessment language are separate.
6. Count-up timer is the MVP default.
7. Benchmarking is language-independent for MVP.
8. External APIs in MVP are basic REST readiness, not full connector implementation.
9. Reports should be stored in S3, with metadata returned through APIs.
10. Audit logging must be applied to all configuration and sensitive business changes.